IAL Physics Unit 2 — The Basics: Charge, Current & Potential Difference | Recall Pack

IAL Physics Unit 2 — The Basics: Charge, Current & Potential Difference

The whole topic in one mental model. Picture each coulomb of charge as a little energy-carrier ("Charlie the Coulomb") riding a hill: the battery sits at the top and an *electromotive force* ε lifts every coulomb up, giving it energy ($\varepsilon = W/Q$). As Charlie rolls round and passes through a component, he gives that energy up — light and heat — and the energy he hands over per coulomb is the *potential difference* $V = W/Q$. *Current* is simply how many coulombs pass a point each second, $I = \Delta Q/\Delta t$, and charge is conserved everywhere: the same current that enters a component leaves it, and at any junction $\sum I_{\text{in}} = \sum I_{\text{out}}$ (Kirchhoff's First Law). Charge itself is quantised, $Q = Ne$ with $e = 1.60 \times 10^{-19}$ C. Two equations carry the whole topic — $I = Q/t$ and $V = W/Q$ — and almost every slip is a unit slip (mA/μA, W vs J, the exponent on e) rather than a physics slip.

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1. Jargon glossary

Term	Student-friendly meaning	Exact physics meaning	Symbols/units	What it is NOT	Lesson example	Phuc/common trap
Electric charge Q	A basic property of matter that comes in two kinds (positive on protons, negative on electrons). Rubbing transfers electrons, so objects become charged; like kinds repel, unlike attract.	A conserved physical property carried by particles; the source of all electrical forces. Total charge in an isolated system is constant (conservation of charge). Comes only in integer multiples of e.	Q; coulombs C	NOT a substance you can use up; NOT continuous — it is quantised; NOT the same as current (charge is the "stuff", current is its *flow rate*).	"Charlie the Coulomb" — each coulomb is a packet of charge riding the hill (Hannah's OneNote; Alex 06:45).	**Phuc trap (06:06):** Phuc gave the circular definition "charge is an object that has some electric charge." Avoid circularity — define charge as the conserved property of protons/electrons that causes electrical forces, NOT "a thing that is charged."

Term	Student-friendly meaning	Exact physics meaning	Symbols/units	What it is NOT	Lesson example	Phuc/common trap
Electric current I	How fast charge flows past a point — coulombs per second. More charge per second = bigger current.	The rate of flow of charge: $I = \Delta Q/\Delta t$. The SI base quantity for electricity; 1 A = 1 C s^{-1}.	I; amperes A $=$ C s^{-1}	NOT "flow of electrons" (loses marks — carriers can be ions too); NOT charge itself; NOT energy flow. The mark scheme wants **"rate of flow of charge."**	Phuc correctly recalled "current is the flow of charged particles" (06:06); refined to *rate of flow of charge*.	**Common trap:** defining current as "flow of electrons" loses the mark — say **"rate of flow of charge"** (ions carry current in electrolytes/gases too).
The coulomb (C)	The unit of charge. One coulomb is the charge that flows in one second when the current is one amp. It is a *big* pile of electrons — about 6.25 billion billion of them.	The SI unit of electric charge; 1 C $= 1$ A · s. Contains 6.25×10^{18} elementary charges.	C; 1 C $= 1$ A s	NOT a single particle; NOT a tiny amount (it's large for lab circuits — we usually use mC, µC, nC).	At 51:55 Phuc asked "is one coulomb like one object?" — Alex: it's a *quantity* (~6.24×10^{18} electrons).	**Phuc trap (51:55):** Phuc thought a coulomb might be a single object. It is a *unit of quantity* — 6.25×10^{18} electrons — which is why a charge of 0.72 C makes perfect sense.
The ampere (A)	The unit of current: one coulomb of charge passing each second.	The SI base unit of electric current; 1 A $= 1$ C s^{-1}. Defined via the force between parallel current-carrying wires.	A; 1 A $= 1$ C/s	NOT a unit of charge (that's the coulomb); NOT energy.	Textbook: "an ampere is quite a large current, so we often use mA and µA."	**Common trap:** mixing up amperes (rate) with coulombs (amount). Units check: A = C/s, so $I = Q/t$ (divide, never multiply).
Elementary charge e	The size of the charge on one electron (or one proton) — the smallest lump of charge there is. Memorise it.	The magnitude of the charge on a single electron/proton: $e = 1.60 \times 10^{-19}$ C. Charge only exists in integer multiples of e (quantisation).	$e = 1.60 \times 10^{-19}$ C	NOT 1.6×10^{-13}; NOT to be written with its sign when *counting* particles (use the magnitude).	Phuc looked it up in the formula booklet: "$e = -1.60 \times 10^{-19}$ C" and asked whether to memorise it (Alex: yes).	**Phuc trap (52:30):** Phuc recalled $e = 1.6 \times 10^{-13}$ (wrong exponent → answer wrong by 10^{6}). Memorise **1.60×10^{-19}** C. Sanity check: $1/e \approx 6 \times 10^{18}$ = electrons in one coulomb.

Term	Student-friendly meaning	Exact physics meaning	Symbols/units	What it is NOT	Lesson example	Phuc/common trap
Charge carrier	A particle that is free to move and so can carry charge round a circuit — free electrons in a metal, ions in a solution or gas.	An electron detached from its atom, or an ion in a conducting medium, that is free to move and thereby constitute a current.	(electrons; ions)	NOT fixed nuclei/atoms; NOT present-but-frozen — they must be *free to move*; NOT always electrons (ions in electrolytes/gases).	Marble-tube analogy (Alex 09:18): push one carrier in, one pops out the far end almost instantly.	**Common trap:** assuming carriers are always electrons. In electrolytes and ionised gases the carriers are **ions** — which is why "flow of charge" beats "flow of electrons."
Quantisation of charge	Charge always comes in whole-number multiples of e — you can't have half an electron's worth.	$Q = Ne$, where N is an integer (number of charge carriers) and e is the elementary charge.	$Q = Ne$; N integer	NOT continuous; NOT "Q can be any value" at the particle level (macroscopically it looks smooth because N is huge).	To count electrons: find $Q = It$, then $N = Q/e$ (Worked Ex. 2).	**Common trap:** using the *signed* value of e when counting carriers. Use the magnitude — the sign only encodes direction, not particle count.
Potential difference (p.d.) V	The energy each coulomb gives up as it passes through a component — energy *out* of the charge, into light/heat/motion. Also called voltage.	The work done (electrical energy converted to other forms) per unit charge moving between two points: $V = W/Q$. Measured in volts.	V; volts V $=$ J C^{-1}	NOT a force; NOT the same as e.m.f. (p.d. = energy *taken from* charge; e.m.f. = energy *given to* charge); NOT current.	"Charlie loses energy at the resistor — same number of Charlies emerge" (hill analogy 06:45). Phuc could not define voltage at 06:40 until the analogy.	**Phuc trap (06:40):** Phuc could not state the definition of p.d. Lock it in: **work done (energy transferred) per unit charge**, $V = W/Q$; the volt $= 1$ J/C.
The volt (V)	One volt means one joule of energy is transferred for every coulomb that passes.	The unit of p.d. and e.m.f.: 1 V $= 1$ J C^{-1}.	V; 1 V $= 1$ J/C	NOT a unit of energy (that's the joule) or charge (coulomb).	Mark-scheme answer for "define the volt": *one joule per coulomb*.	**Common trap:** answering "the volt is the unit of voltage" — circular and zero marks. Say **1 J/C**.

Term	Student-friendly meaning	Exact physics meaning	Symbols/units	What it is NOT	Lesson example	Phuc/common trap
Electromotive force (e.m.f.) ε	The energy a *source* (battery, generator) gives to each coulomb — energy *into* the charge. It is not actually a force despite the name.	The energy converted *into* electrical energy per unit charge by a source: $\varepsilon = W/Q$. Same units as p.d. (volts), opposite energy direction.	ε; volts V $=$ J C^{-1}	NOT a force (it's an energy-per-charge); NOT the same as p.d. (e.m.f. = energy *supplied to* charge; p.d. = energy *taken from* charge).	Source guide: "If there were no internal resistance, e.m.f. would equal the terminal p.d." (Topic 2.4 conceptual).	**Common trap:** calling e.m.f. a "force." It is energy transfer per coulomb (J/C = V). Distinguish from p.d. by *energy direction*: ε into charge, V out of charge.
Conventional current	The direction we *pretend* current flows: out of the + terminal, round the circuit, into the − terminal. A historical convention chosen before electrons were known.	The flow direction of (notional) positive charge: positive terminal → external circuit → negative terminal. Used in all circuit analysis unless electron flow is specifically asked.	direction: $+ \to -$	NOT the electron-flow direction (it's the opposite); NOT "wrong" — it's the agreed standard.	Source diagram 4.2: conventional current $+ \to -$, electron flow $- \to +$, drawn as opposite arrows.	**Common trap:** writing conventional current as $- \to +$. Conventional current is **positive terminal → negative terminal**; electrons go the *other* way.
Electron flow	The way the electrons *actually* move in a metal wire: out of the − terminal toward the + terminal — opposite to conventional current.	The drift direction of free electrons in a metallic conductor: negative terminal → external circuit → positive terminal.	direction: $- \to +$	NOT the conventional current direction; NOT the direction used by default in calculations.	Source: "electrons flow from negative to positive; conventional current is opposite."	**Common trap:** when asked "conventional vs electron flow," state **both directions explicitly** and note electrons go − to +. Don't assume they're the same.
Ammeter	The meter that measures current. Goes *in line* (series) with the component so the same current passes through it.	A current-measuring instrument of *low* resistance, connected in series so it does not disturb the current it measures.	(series; low R)	NOT connected in parallel; NOT high-resistance; does NOT measure voltage.	Source diagram 4.1: "Ammeter in series; Voltmeter in parallel."	**Common trap:** putting the ammeter in parallel. Ammeter = **series** (so the same current flows through ammeter and component); it has *low* resistance.

Term	Student-friendly meaning	Exact physics meaning	Symbols/units	What it is NOT	Lesson example	Phuc/common trap
Voltmeter	The meter that measures p.d. Goes *across* (parallel with) the component, and must barely sip current so it doesn't change the circuit.	A p.d.-measuring instrument of *very high* resistance, connected in parallel across a component so it draws negligible current.	(parallel; high R)	NOT in series; NOT low-resistance; does NOT measure current.	Textbook: a 20 V digital voltmeter might have 10 MΩ resistance, drawing tiny current i.	**Common trap:** forgetting *why* it's high-resistance — "so it draws negligible current and does not disturb the circuit." Voltmeter = **parallel**, high R.
Kirchhoff's First Law (KCL)	At any junction, all the current flowing in equals all the current flowing out — charge can't pile up or vanish.	The algebraic sum of currents at a junction is zero: $\sum I_{\text{in}} = \sum I_{\text{out}}$. A direct consequence of conservation of charge; applies at *every* junction in *every* circuit.	$\sum I_{\text{in}} = \sum I_{\text{out}}$; A	NOT a rule only for parallel circuits; NOT a special-topology rule — it's universal charge conservation.	Source diagram 4.3: $I = I_1 + I_2 \Rightarrow 5$ mA $= 2 + 3$ mA.	**Common trap:** stating KCL without the *reason*. Examiners want "**because charge is conserved / cannot accumulate at a junction.**" Also: it applies everywhere, not only in parallel.
Conservation of charge	Charge is never created or destroyed. In a series circuit the current is the same everywhere; a resistor passes charge straight through.	A fundamental law: total charge in an isolated system is constant. Underlies KCL and the equal-current-in-series rule.	(principle)	NOT energy conservation (separate law); does NOT mean a resistor "stores" charge.	"Is it the same Charlie?" (09:18) — and "does a resistor store electrons?" Answer: no.	**Phuc trap (09:18):** Phuc wondered whether charge accumulates / whether it's the same electron throughout. A resistor **converts energy, not stores charge** — same current flows in as out.

Term	Student-friendly meaning	Exact physics meaning	Symbols/units	What it is NOT	Lesson example	Phuc/common trap
Drift velocity (conceptual)	Electrons themselves crawl along very slowly (mm/s), yet the circuit lights up instantly because the *push* (electric field) travels near light-speed.	The slow net drift speed of charge carriers under an applied field; the field itself propagates at $\sim c$, so the current effect is near-instant even though carriers are slow.	(mm s^{-1}, conceptual)	NOT the speed at which the circuit "turns on"; NOT near light-speed (that's the field, not the electrons).	Marble-tube analogy (09:18): push one marble in, one pops out instantly; each moved only a little.	**Common trap:** confusing the (fast) field propagation with the (slow) electron drift. Electrons drift slowly; the *field* propagates fast — that's why the lamp lights immediately.

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2. Formula / case table

Quantity / case	Formula	Symbol meanings	Units	When to use	Rearrangements	Trap	Mini worked example
Electric current	$I = \dfrac{\Delta Q}{\Delta t}$	I = current; ΔQ = charge passing; Δt = time	I in A; Q in C; t in s	Finding current, charge, or time from any two of them; the definition of current.	$\Delta Q = I\Delta t$; $\Delta t = \dfrac{\Delta Q}{I}$	**Phuc trap (15:00):** divide, don't multiply — Phuc wrote $I = Q \times t$ then self-corrected. Units fix it: A = C/s $\Rightarrow I = Q/t$.	$I = 0.5$ A for $5\,\text{min}\,30\,\text{s} = 330$ s: $Q = I\Delta t = 0.5 \times 330 = 165$ C.
Charge from current & time	$\Delta Q = I\,\Delta t$	ΔQ = total charge; I = current; Δt = time	C, A, s	"How much charge flows in this time?" Always convert min-utes→seconds, mA/μA→A first.	$I = \dfrac{\Delta Q}{\Delta t}$; $\Delta t = \dfrac{\Delta Q}{I}$	Forgetting to convert time to seconds, or current to amps, before substituting.	Filament lamp: $I = 250$ mA $= 0.250$ A, $\Delta t = 1$ h $= 3600$ s $\Rightarrow Q = 0.250 \times 3600 = 900$ C.

Quantity / case	Formula	Symbol meanings	Units	When to use	Rearrangements	Trap	Mini worked example
Charge quantisation	$Q = Ne$	Q = total charge; N = number of carriers (integer); $e = 1.60{\times}10^{-19}$ C	C; N dimensionless	Counting electrons/ions, or finding total charge from a carrier count.	$N = \dfrac{Q}{e}$; $e = \dfrac{Q}{N}$	**Phuc trap (52:30):** use $e = 1.60{\times}10^{-19}$ (NOT 10^{-13}); use the **magnitude** of e when counting (ignore the sign).	$Q = 0.72$ C $\Rightarrow N = 0.72/(1.60 \times 10^{-19}) = 4.5 \times 10^{18}$ electrons.
Count electrons from current	$N = \dfrac{Q}{e} = \dfrac{I\,\Delta t}{e}$	combine $Q = I\Delta t$ then $N = Q/e$	dimensionless	"How many electrons pass in this time?" — the classic two-step.	$Q = Ne$; $I = \dfrac{Ne}{\Delta t}$	**Phuc trap (15:03):** µA = 10^{-6}, not 10^{-3} — convert the current *first*.	Beam $I = 1.0\,\mu$A $= 1.0 \times 10^{-6}$ A, $t = 600$ s: $Q = 6.0 \times 10^{-4}$ C, $N = 6.0 \times 10^{-4}/1.60 \times 10^{-19} = 3.75 \times 10^{15}$.
Potential difference	$V = \dfrac{W}{Q}$	V = p.d.; W = electrical energy converted to other forms; Q = charge	V, J, C	Defining/finding p.d.; energy delivered *by* charge to a component.	$W = VQ$; $Q = \dfrac{W}{V}$	**Phuc trap (06:40):** know the definition — *work done per unit charge*. Energy flows *out* of the charge here (≠ e.m.f.).	Element: $W = 540$ kJ, $Q = 2250$ C $\Rightarrow V = 540 \times 10^3/2250 = 240$ V.
E.m.f. (definition)	$\varepsilon = \dfrac{W}{Q}$	ε = e.m.f.; W = energy converted *into* electrical form by source; Q = charge	V, J, C	Energy a source *supplies* per coulomb; defining/explaining e.m.f.	$W = \varepsilon Q$	Same algebra as V but energy goes **into** the charge (source) vs out (component). E.m.f. is not a force.	Battery converts $W = 12150$ J to $Q = 1350$ C $\Rightarrow \varepsilon = 12150/1350 = 9.0$ V.

Quantity / case	Formula	Symbol meanings	Units	When to use	Rearrangements	Trap	Mini worked example
Energy from p.d. & charge	$W = VQ$	W = energy transferred; V = p.d.; Q = charge	J, V, C	Energy delivered when charge Q passes through p.d. V.	$V = \frac{W}{Q}$; $Q = \frac{W}{V}$	Don't stop at $V = IR$ when the question wants *energy* — you still need $W = VQ$ or $W = Pt$.	$\varepsilon = 9$ V battery delivers $Q = 1350$ C $\Rightarrow W = 9 \times 1350 = 12150$ J.
Electrical power	$P = VI$	P = power; V = p.d.; I = current	W, V, A	Power developed in a component/source when both V and I are known.	$V = \frac{P}{I}$; $I = \frac{P}{V}$	**Phuc trap (47:39):** $P = VI$ and $P = W/t$ give the *same* answer — don't think they differ.	Battery $V = 9$ V, $I = 0.5$ A $\Rightarrow P = 9 \times 0.5 = 4.5$ W.
Power from energy & time	$P = \frac{W}{t}$	P = power; W = energy; t = time	W, J, s	When you know energy and time directly.	$W = Pt$; $t = \frac{W}{P}$	**Phuc trap (47:07):** a watt is a joule *per second* (1 W = 1 J/s) — don't confuse W and J.	$W = 12150$ J over $t = 2700$ s $\Rightarrow P = 12150/2700 = 4.5$ W (matches VI).
Energy from power & time	$W = Pt$	W = energy; P = power; t = time	J, W, s	The *fastest* route to energy when P and t are known — skip Q entirely.	$P = \frac{W}{t}$; $t = \frac{W}{P}$	**Phuc trap (31:03):** Phuc went the long way (I→Q→$W = VQ$). If you know P and t, use $W = Pt$ directly.	$P = 800$ W appliance for $t = 60$ s $\Rightarrow W = 800 \times 60 = 48000$ J.
Ohm's law	$V = IR$	V = p.d.; I = current; R = resistance	V, A, Ω	Linking p.d., current and resistance (next topic, but appears here).	$I = \frac{V}{R}$; $R = \frac{V}{I}$	**Phuc trap (54:16):** Phuc hesitated recalling it. Use $R = V/I$ — **never** $R = IV$.	$V = 12$ V across $I = 40$ mA $= 0.040$ A $\Rightarrow R = 12/0.040 = 300\ \Omega$.

Quantity / case	Formula	Symbol meanings	Units	When to use	Rearrangements	Trap	Mini worked example
Kirchhoff's First Law	$\sum I_{\text{in}} = \sum I_{\text{out}}$	currents entering a junction = currents leaving	A	Junctions in parallel circuits; checking series current is constant.	(additive: $I = I_1 + I_2 + \ldots$)	State the **reason** (charge conservation) for the explanation mark; it applies at *every* junction, not only in parallel.	2.0 A and 3.5 A enter a junction $\Rightarrow I_{\text{out}} = 2.0 + 3.5 = 5.5$ A.
Prefix conversions	m $= 10^{-3},\ \mu = 10^{-6},$ n $= 10^{-9}$	milli / micro / nano multipliers	(multipliers)	Converting mA, μA, nA → A (and mC, μC, nC → C) **before** substituting.	multiply by the power of ten and substitute in SI.	**Phuc trap (15:03):** μ (micro) $= 10^{-6}$, NOT 10^{-3} — confusing them is wrong by a factor of 1000.	40 mA $= 40 \times 10^{-3}$ A $= 0.040$ A; $1.0\,\mu$A $= 1.0 \times 10^{-6}$ A.

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3. Energy vs charge — one-glance contrast (the question Phuc kept asking)

Quantity	Formula	"per second" form	What flows / is measured	Unit	Plain-English
Charge Q	$Q = It$	$I = Q/t$ = rate of flow of **charge**	the amount of charge (carriers)	C	How many coulombs have passed.
Current I	$I = \Delta Q/\Delta t$	(is itself the per-second rate)	rate of charge flow	A = C/s	How fast charge flows past a point.
Energy W	$W = VQ = Pt$	$P = W/t$ = rate of transfer of **energy**	the energy carried/delivered	J	How much energy is transferred.
Power P	$P = VI = W/t$	(is itself the per-second rate)	rate of energy transfer	W = J/s	How fast energy is transferred.

Phuc's own synthesis (confirmed correct, ~33 min): "$I = Q/t$ is the number of objects [coulombs] potentially carrying energy; $P = E/t$ is the actual energy transferred per second." Current counts the carriers; power counts the energy.

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4. Big picture (last-minute recall)

- **Two master equations:** $I = \dfrac{Q}{t}$ (divide, never $Q \times t$) and $V = \dfrac{W}{Q}$. Almost everything else is a rearrangement.
- **Current = rate of flow of charge** (mark-scheme wording) — *not* "flow of electrons" (ions count too).
- $e = 1.60 \times 10^{-19}$ **C** — memorise the exponent (10^{-19}, not 10^{-13}). Counting electrons: $N = Q/e$, use the magnitude.
- **Prefixes:** $\mu = 10^{-6}$, m $= 10^{-3}$, n $= 10^{-9}$. Convert mA/μA → A and minutes → seconds *before* substituting.
- **Units discipline:** A = C/s, V = J/C, W = J/s. Watt is energy *per second*; joule is energy. Don't swap W and J.
- **e.m.f. vs p.d.:** ε = energy *into* charge (source); V = energy *out of* charge (component). Both in volts; e.m.f. is not a force.
- **Conventional current $+ \to -$; electron flow $- \to +$** — opposite directions; state both when asked.
- **Ammeter in series (low R); voltmeter in parallel (high R, negligible current).**
- **Kirchhoff's 1st Law:** $\sum I_{\text{in}} = \sum I_{\text{out}}$ at *every* junction — because **charge is conserved**. A resistor converts energy, it does not store charge: same current in as out.
- **Fastest energy route:** if you know P and t, use $W = Pt$ directly — don't detour through Q.